

Feature



New Nehalem

The new i7 975 XE is based around the latest DO stepping and is clocked at 3.33 GHz

DDR3

DDR3 is already clocking well over 2 GHz and promises of twice the speed of DDR2 have been made



SATA 3.0

Although it's still in the works, SATA 3.0 with speeds of up to 600 MB/s is promised soon



So we wrote to AMD asking for the newest 3.2 GHz Phenom 2 955 BE. The BE suffix signifies a black edition. This means that the CPU multiplier is unlocked – a boon for overclockers, especially if the motherboard does not have much potential for overclocking its FSB/HTT. We also asked for a motherboard based on the new AM3 platform and we were sent a Gigabyte GA-MA790FX-UD5P based on the 790FX.

This is a very well laid out and solidly built board, with an all solid-state capacitor design.

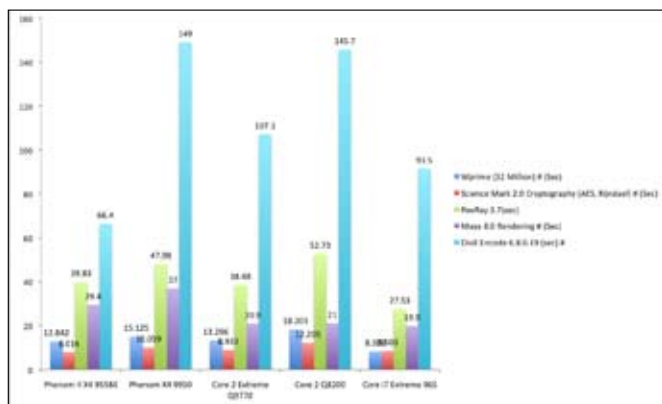
There's even a small heatsink provided for the additional SATA controller that provides four SATA ports in addition to the SB750s (Southbridge) six. One of the talked-about inclusions in terms of features was EC AOD-ACC – an acronym for "Embedded Controller for AMD OverDrive-Advanced Clock Calibration" that is part of the new Southbridge and allows AMD Black Edition CPUs to be overclocked. The BIOS is rich with overclocking options, although you do need to know exactly what you're doing. AMD also sent us DDR3

memory in the form of two sticks of Corsair Dominator memory rated at 1600 MHz at timings of 9-9-9-24 and a voltage of 1.65. We were also sent an ATI HD 4890 graphics card, although for all the graphics related tests we stuck to our reference NVIDIA GTX 285 solution for the sake of uniformity. Also in the box, was a small but efficient CPU cooler.

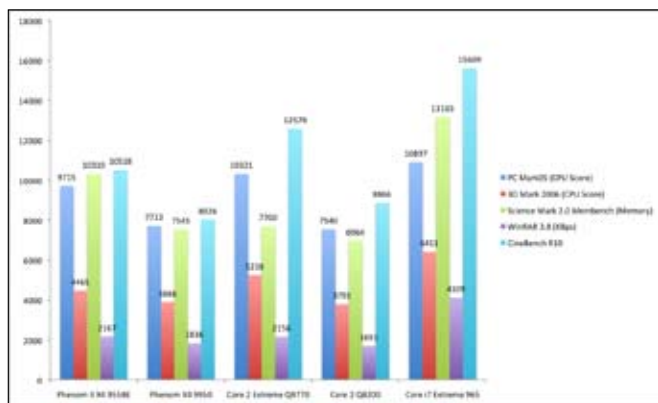
Once everything was setup we noticed how cool the processor was running. With the room at an ambient 25 degrees, the CPU went to 37 degrees and stayed there for a good while. Upon stress-benchmarking, we noticed a rise of not more than nine degrees and this testifies to the fact that the 45nm Phenom 2 is a lot cooler than the 65nm

Phenom processors, even at a much higher clock speed. For the record we did manage to overclock this processor to a stable 4.1 GHz using air-cooling, and although the temperature soared to 56 degrees it was obvious that with a better cooling solution we could crack 4.8 GHz easily, however we didn't push it. So AMD's Phenom 2 overclocks well, but how did it do on stock?

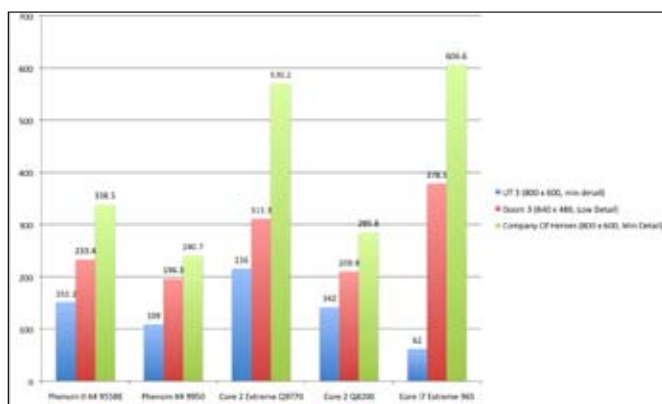
The Phenom 2 955 BE is just about available here and its pricing is not stable at the moment. We expect it to fall to around the Rs. 14,000 mark but the best we could find was Rs. 15,000 making it costlier than the Phenom X4 9950, but cheaper than the Rs. 17,000 Intel Core 2 Q9550 (2.83 GHz). As the benchmarks show (note the bold figures indicate the best performer) this CPU actually takes the lead in a couple of benchmarks and this is astounding given the company it's in. On the whole, it's still much slower than the mighty Core i7 965 Extreme and quite a bit behind the Core 2 Extreme Q9770, but it trounces the Core 2 Duo Q8200, something the Phenom X4 9950 wasn't able to achieve; this proves AMD is finally on the right track. The optimisations to the Barcelona core, the extra L3 cache, the improved clock speeds and its excellent thermal characteristics are testament to a great product. It is however, nearly twice the price of the Q8200 and we could not get our hands on either a Q9450 (2.66 GHz) or a Q9550 (2.83 GHz) both of which are similarly priced and therefore more comparable. Our intuition tells us that this CPU will come very close to the Q9550 and perhaps beat the Q9450 in a good many benchmarks. This is high praise for a company that six months back seemed to have no hope of providing even cursory competition to Intel. AMD's fortunes have far from changed, after all Intel has promised 32nm Nehalem variants in early 2010, but the Phenom 2 marks the beginning of renewed hope for the green team. After all, everybody likes a good fight. **4**



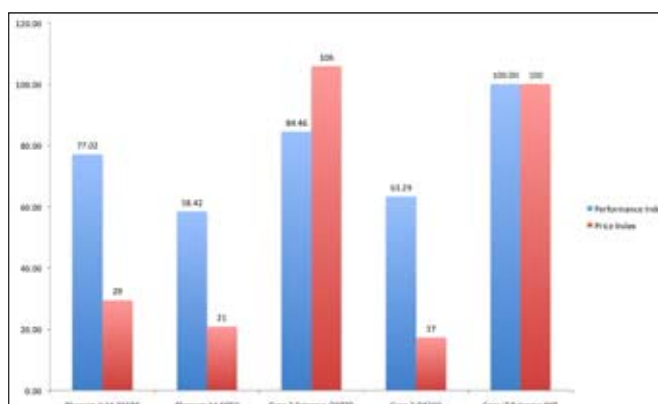
All these scores are in seconds, so the lesser the better



Synthetic and real world tests



The all important games. Note the steady improvement over the 1st generation Phenom



With 77 per cent of Core i7's performance at 29 per cent of its price, the Phenom 2 makes a solid splash as a really fast, value CPU